Practice workbook · Interpret antibacterial culture investigations

Petri dish and growth-graph investigations of antibiotics or bacterial growth: comparing inhibition zones or counts, variables and safe incubation.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: HBI-ANTIBACTERIAL-CULTURE-INVESTIGATION

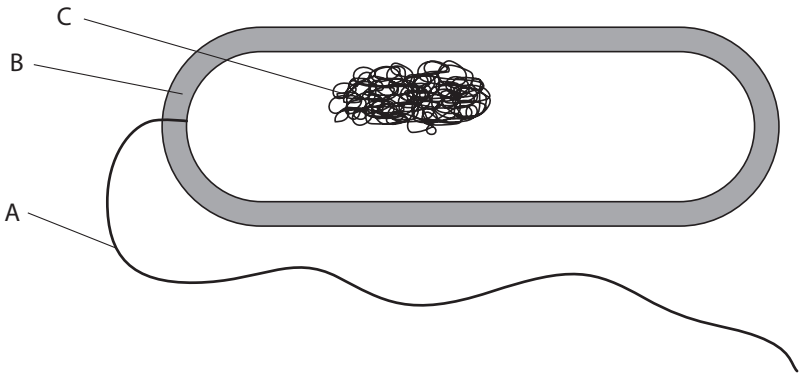
1. Question 8 · part b.ii · 3 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

8 The diagram shows a bacterial cell.



(a) (i) Name the parts labelled A, B and C.

(3)

A .....

B .....

C .....

(ii) Some types of bacteria cause disease.

Give the name of organisms that cause disease.

(1)

.....

OLevels.com SCHOOL

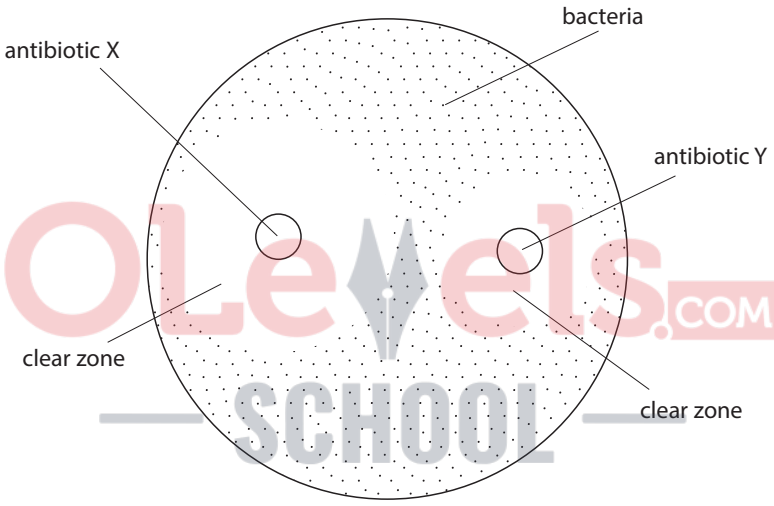
1. Question 8 · part b.ii · 3 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b.ii only.

(b) A student uses this method to investigate whether antibiotic X is more effective than antibiotic Y.

- grow bacteria on nutrient agar in a Petri dish
- soak one disc of filter paper in antibiotic X and another disc in antibiotic Y
- place each disc on the agar in the Petri dish
- seal the Petri dish and leave for three days



(i) Describe what is meant by the term **antibiotic**.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 2 5 8 6 A 0 1 8 2 0

1. Question 8 · part b.ii · 3 marks

June 2023 Paper 1R

Worked-example retry · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

(ii) Explain which antibiotic, X or Y, is more effective against the bacteria in the Petri dish.

(3)

(iii) Explain a suitable control that could be used in this investigation.

(3)

(Total for Question 8 = 12 marks)

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 8 · part b.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 7 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

7 Various diseases are caused by microorganisms. Some diseases can be treated by using antibiotics.

(a) (i) Which of these is the name given to any microorganisms that cause disease? **(1)**

- ☒ **A** bacteria
- ☒ **B** fungi
- ☒ **C** pathogens
- ☒ **D** protozoa

(ii) Explain why it is not possible to treat Ebola using antibiotics. **(2)**

.....

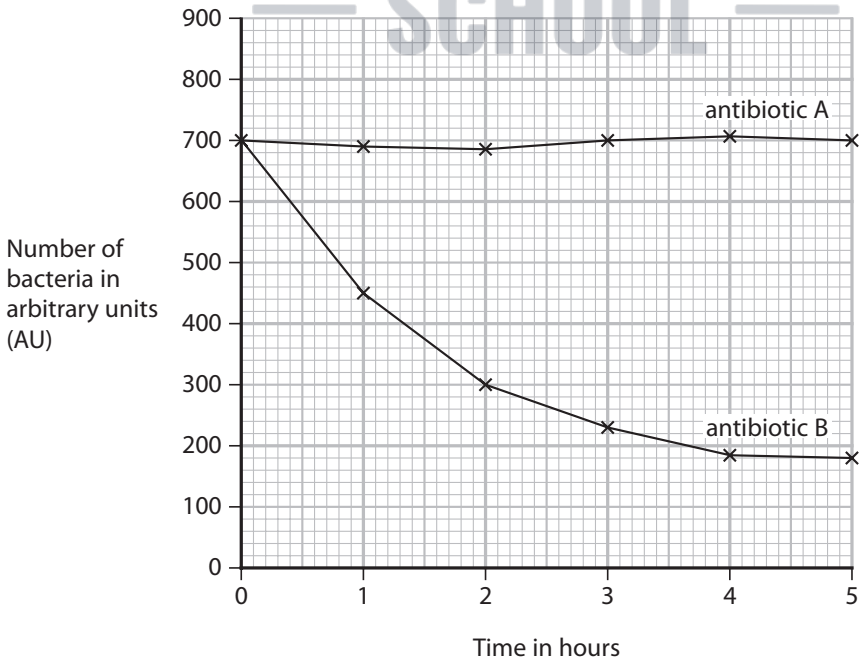
.....

.....

.....

OLevels.com SCHOOL

(b) The graph shows the effect of two different antibiotics on the growth of a population of bacteria.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

2. Question 7 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(i) Calculate the mean rate of change of the bacterial population for antibiotic B during the five hours.

Give your answer to 2 significant figures.

(3)

mean rate of change = ... AU per hour

(ii) A bactericidal antibiotic kills bacteria.

A bacteriostatic antibiotic prevents bacteria from reproducing.

Explain which antibiotic is bactericidal.

Use data from the graph to help your answer.

(2)

OLevels.COM — SCHOOL —

.....

.....

.....

.....

2. Question 7 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

(iii) Discuss which antibiotic would be better to use to treat any disease caused by the bacteria.

(3)

DO NOT WRITE IN THIS AREA

DO NOT V

OLevels.com SCHOOL

(Total for Question 7 = 11 marks)

Working and answer

3. Question 7 · part b.iii · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.iii only.

7 Various diseases are caused by microorganisms. Some diseases can be treated by using antibiotics.

(a) (i) Which of these is the name given to any microorganisms that cause disease? **(1)**

- ☒ **A** bacteria
- ☒ **B** fungi
- ☒ **C** pathogens
- ☒ **D** protozoa

(ii) Explain why it is not possible to treat Ebola using antibiotics. **(2)**

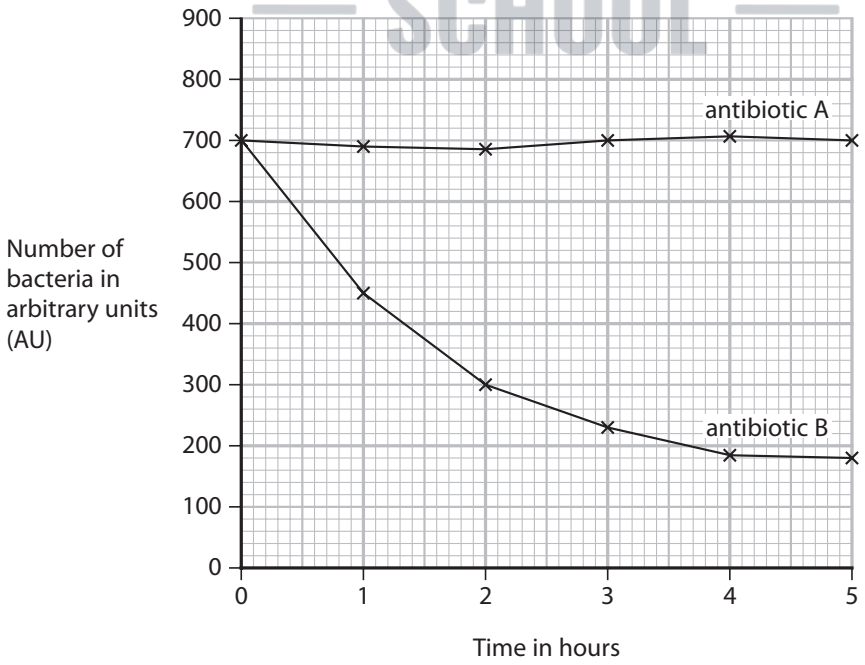
.....

.....

.....

.....

(b) The graph shows the effect of two different antibiotics on the growth of a population of bacteria.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

3. Question 7 · part b.iii · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(i) Calculate the mean rate of change of the bacterial population for antibiotic B during the five hours.

Give your answer to 2 significant figures.

(3)

mean rate of change = ... AU per hour

(ii) A bactericidal antibiotic kills bacteria.

A bacteriostatic antibiotic prevents bacteria from reproducing.

Explain which antibiotic is bactericidal.

Use data from the graph to help your answer.

(2)

OLevels.COM — SCHOOL —

3. Question 7 · part b.iii · 3 marks

November 2021 Paper 2

Further practice · Attempt part b.iii only.

(iii) Discuss which antibiotic would be better to use to treat any disease caused by the bacteria.

(3)

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 7 = 11 marks)

Working and answer

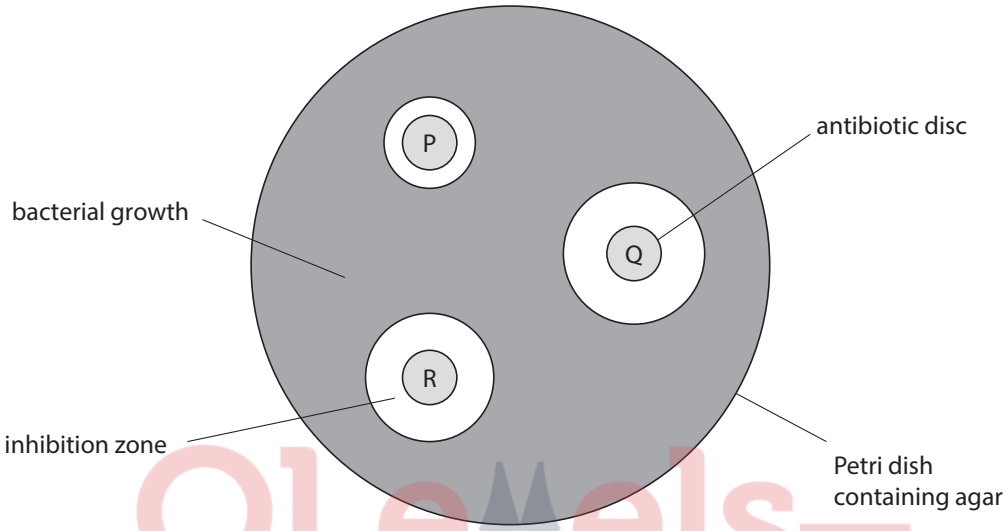
4. Question 5 · part a.iii · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.iii only.

5 (a) A student investigates the action of three different antibiotics, P, Q and R, on the growth of bacteria.

The diagram shows the results of the investigation.



(i) Which is the independent variable in this investigation?

(1)

- ☒ **A** the mass of bacteria added
- ☒ **B** the size of the inhibition zone
- ☒ **C** the type of antibiotic added
- ☒ **D** the mass of agar in the Petri dish

(ii) Give two procedures that the student should use to reduce the risk of contamination by unwanted bacteria.

(2)

1 .....

.....

.....

2 .....

.....

.....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 4 6 3 0 A 0 1 4 2 4

4. Question 5 · part a.iii · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.iii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

IN THIS AREA

(iii) Explain which is the most effective antibiotic.

(2)

(iv) The student concludes the bacteria are resistant to antibiotic P.

Explain how antibiotic resistance occurs in bacteria.

(4)

OLevels.com SCHOOL

(b) Decomposers are microorganisms such as bacteria and fungi that do not cause disease.

Describe the role of these microorganisms in the environment.

(2)

(Total for Question 5 = 11 marks)

4. Question 5 · part a.iii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 7 · part b.ii · 4 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

7 Garlic plants contain a substance called allicin. This substance is thought to have antibacterial properties.

A student investigates the antibacterial effect of garlic.

This is the student's method.

- label three test tubes A, B and C
- add 10 cm^3 of milk into each of the tubes
- add a sample of the bacterium *E. coli* into tubes B and C using aseptic techniques
- crush the garlic in water to form garlic juice
- add 1 cm^3 of garlic juice to tube C
- shake all three tubes and leave to incubate for two hours
- place a sample from tube A into the centre of Petri dish A, from tube B into the centre of Petri dish B and from tube C into the centre of Petri dish C
- incubate the three Petri dishes for five days
- each day measure the diameter of any bacterial growth that has occurred

The table shows the student's results.

Petri dish	Diameter of bacterial growth in mm				
	day 1	day 2	day 3	day 4	day 5
A	0	1.7	3.0	4.6	7.1
B	0	4.2	8.4	15.1	36.5
C	0	0	0	0	0

(a) (i) State three control variables in the student's investigation.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

5. Question 7 · part b.ii · 4 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

(ii) State the dependent variable in the student's investigation.

(1)

.....

(b) (i) Calculate the mean daily increase in the diameter of the bacterial growth for dish A and dish B.

(2)

mean daily increase for dish A ..... mm/day

mean daily increase for dish B ..... mm/day

DO NOT WRITE IN THIS AREA

(ii) Explain the difference between the mean daily increases in the diameter of the bacterial growth for dish A and dish B.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS AREA

(c) Explain what is meant by an aseptic techniques for the transfer of bacteria.

(2)

.....

.....

.....

.....

O Levels .com — SCHOOL —

5. Question 7 · part b.ii · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL